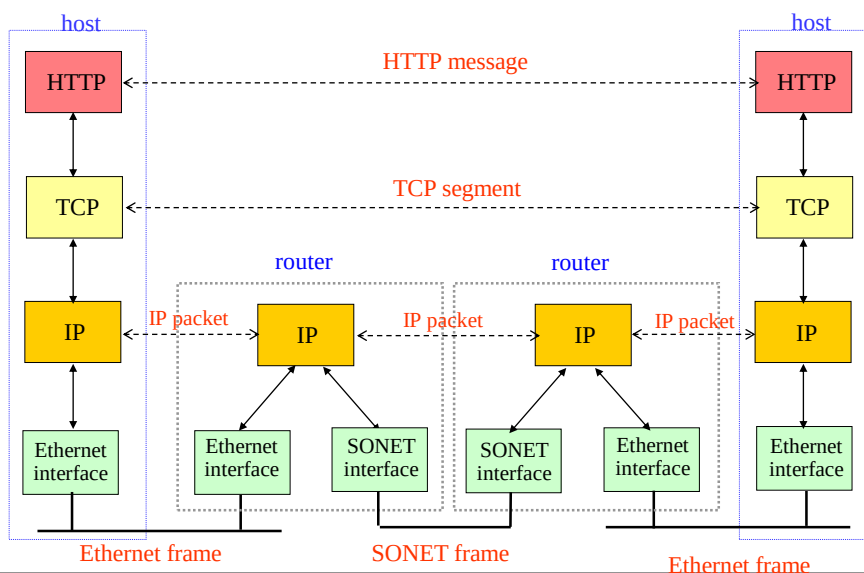


05.04 Networking Devices

Goals of Today's Lecture

- Devices that shuttle data at different layers
 - Repeaters and hubs
 - Bridges and switches
 - Routers
- Switch protocols and mechanisms
 - Dedicated access and full-duplex transfers
 - Cut-through switching
 - Self learning of the switch table
 - Spanning trees

Message, Segment, Packet, and Frame



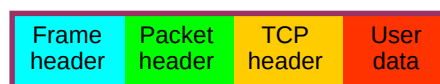
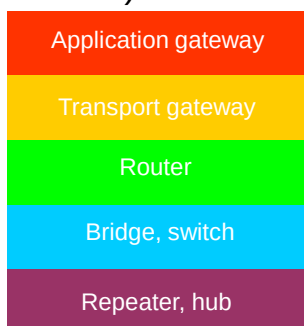
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Shuttling Data at Different Layers

- Different devices switch different things
 - Network layer: packets (routers)
 - Link layer: frames (bridges and switches)
 - Physical layer: electrical signals (repeaters and hubs)



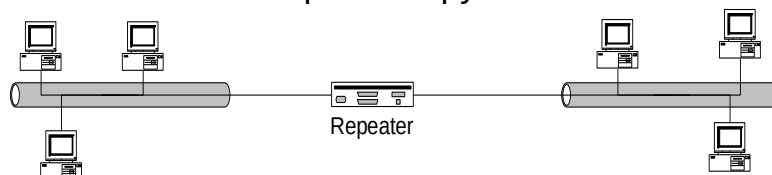
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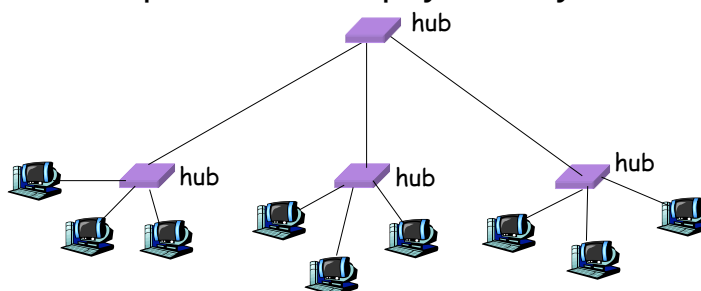
Physical Layer: Repeaters

- Distance limitation in local-area networks
 - Electrical signal becomes weaker as it travels
 - Imposes a limit on the length of a LAN
- Repeaters join LANs together
 - Analog electronic device
 - Continuously monitors electrical signals on each LAN
 - Transmits an amplified copy



Physical Layer: Hubs

- Joins multiple input lines electrically
 - Designed to hold multiple line cards
 - Do not necessarily amplify the signal
- Very similar to repeaters
 - Also operates at the physical layer

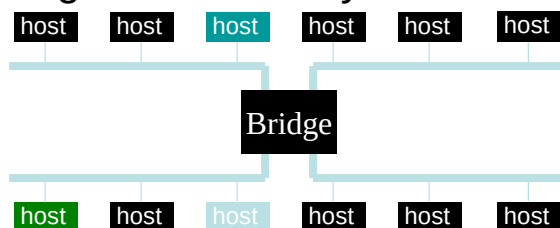


Limitations of Repeaters and Hubs

- One large shared link
 - Each bit is sent everywhere
 - So, aggregate throughput is limited
 - E.g., three departments each get 10 Mbps independently
 - ... and then connect via a hub and must share 10 Mbps
- Cannot support multiple LAN technologies
 - Does not buffer or interpret frames
 - So, can't interconnect between different rates or formats
 - E.g., 10 Mbps Ethernet and 100 Mbps Ethernet
- Limitations on maximum nodes and distances
 - Shared medium imposes length limits (see next lecture)
 - E.g., cannot go beyond 2500 meters on Ethernet

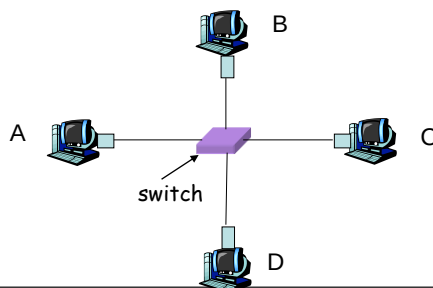
Link Layer: Bridges

- Connects two or more LANs at the link layer
 - Extracts destination address from the frame
 - Looks up the destination in a table
 - Forwards the frame to the appropriate LAN segment
- Each segment can carry its own traffic



Link Layer: Switches

- Typically connects individual computers
 - A switch is essentially the same as a bridge
 - ... though typically used to connect hosts, not LANs
- Like bridges, support concurrent communication
 - Host A can talk to C, while B talks to D

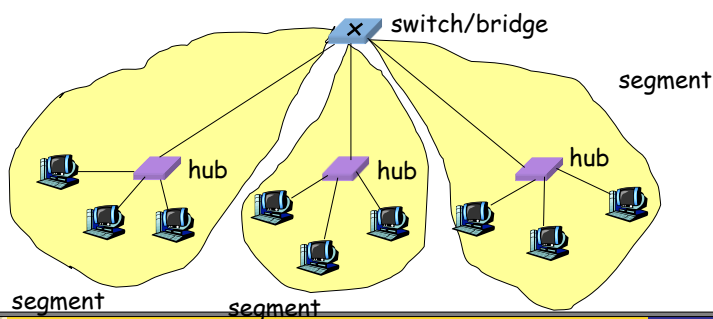


Dedicated Access and Full Duplex

- Dedicated access
 - Host has direct connection to the switch
 - ... rather than a shared LAN connection
- Full duplex
 - Each connection can send in both directions
 - Host sending to switch, and host receiving from switch
 - E.g., in 10BaseT and 100Base T
- Completely supports concurrent transmissions
 - Each connection is a bidirectional point-to-point link

Bridges/Switches: Traffic Isolation

- Switch breaks subnet into LAN segments
- Switch filters packets
 - Frame only forwarded to the necessary segments
 - Segments can support separate transmissions



Advantages Over Hubs/Repeaters

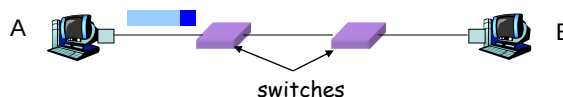
- Only forwards frames as needed
 - Filters frames to avoid unnecessary load on segments
 - Sends frames only to segments that need to see them
- Extends the geographic span of the network
 - Separate segments allow longer distances
- Improves privacy by limiting scope of frames
 - Hosts can “snoop” the traffic traversing their segment
 - ... but not all the rest of the traffic
- Can join segments using different technologies

Disadvantages Over Hubs/Repeaters

- Delay in forwarding frames
 - Bridge/switch must receive and parse the frame
 - ... and perform a look-up to decide where to forward
 - Storing and forwarding the packet introduces delay
 - Solution: cut-through switching
- Need to learn where to forward frames
 - Bridge/switch needs to construct a forwarding table
 - Ideally, without intervention from network administrators
 - Solution: self-learning
- Higher cost
 - More complicated devices that cost more money

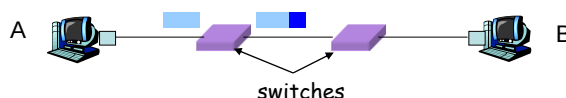
Motivation For Cut-Through Switching

- Buffering a frame takes time
 - Suppose L is the length of the frame
 - And R is the transmission rate of the links
 - Then, receiving the frame takes L/R time units
- Buffering delay can be a high fraction of total delay
 - Propagation delay is small over short distances
 - Making buffering delay a large fraction of total
 - Analogy: large group walking through NYC



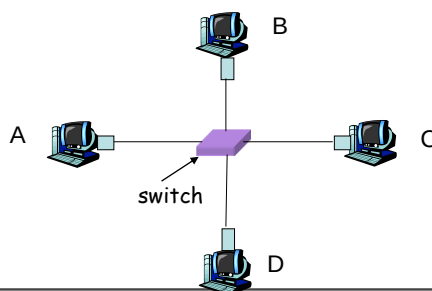
Cut-Through Switching

- Start transmitting as soon as possible
 - Inspect the frame header and do the look-up
 - If outgoing link is idle, start forwarding the frame
- Overlapping transmissions
 - Transmit the head of the packet via the outgoing link
 - ... while still receiving the tail via the incoming link
 - Analogy: different folks crossing different intersections



Motivation For Self Learning

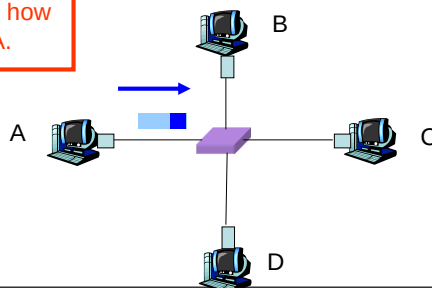
- Switches forward frames selectively
 - Forward frames only on segments that need them
- Switch table
 - Maps destination MAC address to outgoing interface
 - Goal: construct the switch table automatically



Self Learning: Building the Table

- When a frame arrives
 - Inspect the *source* MAC address
 - Associate the address with the *incoming* interface
 - Store the mapping in the switch table
 - Use a time-to-live field to eventually forget the mapping

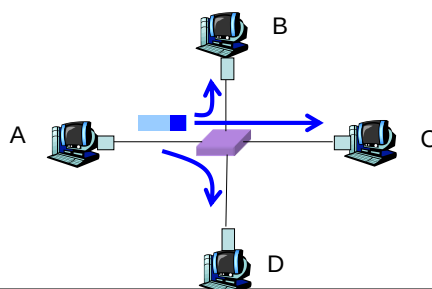
Switch learns how to reach A.



Self Learning: Handling Misses

- When frame arrives with unfamiliar destination
 - Forward the frame out all of the interfaces
 - ... except for the one where the frame arrived
 - Hopefully, this case won't happen very often

When in doubt, shout!



Switch Filtering/Forwarding

When switch receives a frame:

index switch table using MAC dest address

if entry found for destination

then{

if dest on segment from which frame arrived

then drop the frame

else forward the frame on interface indicated

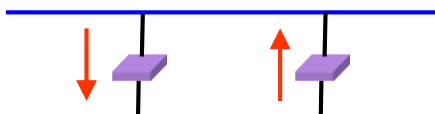
}

else flood

forward on all but the interface
on which the frame arrived

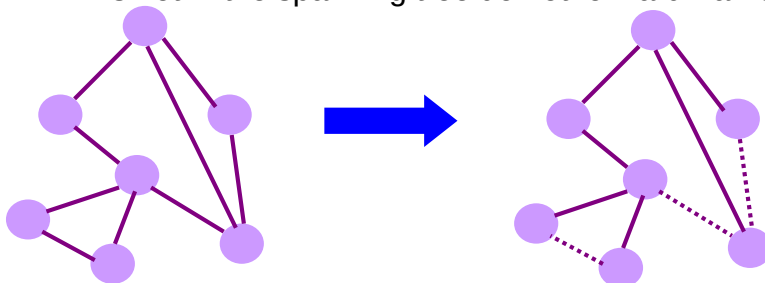
Flooding Can Lead to Loops

- Switches sometimes need to broadcast frames
 - Upon receiving a frame with an unfamiliar destination
 - Upon receiving a frame sent to the broadcast address
- Broadcasting is implemented by flooding
 - Transmitting frame out every interface
 - ... except the one where the frame arrived
- Flooding can lead to forwarding loops
 - E.g., if the network contains a cycle of switches
 - Either accidentally, or by design for higher reliability



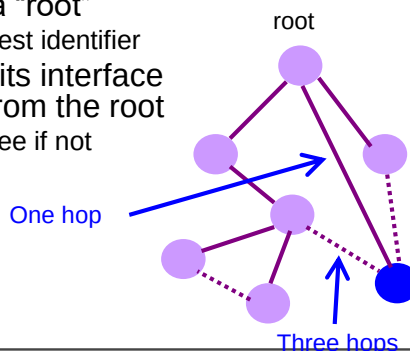
Solution: Spanning Trees

- Ensure the topology has no loops
 - Avoid using some of the links when flooding
 - ... to avoid forming a loop
- Spanning tree
 - Sub-graph that covers all vertices but contains no cycles
 - Links not in the spanning tree do not forward frames



Constructing a Spanning Tree

- Need a distributed algorithm
 - Switches cooperate to build the spanning tree
 - ... and adapt automatically when failures occur
- Key ingredients of the algorithm
 - Switches need to elect a “root”
 - The switch with the smallest identifier
 - Each switch identifies if its interface is on the shortest path from the root
 - And it exclude from the tree if not
 - Messages (Y, d, X)
 - From node X
 - Claiming Y is the root
 - And the distance is d

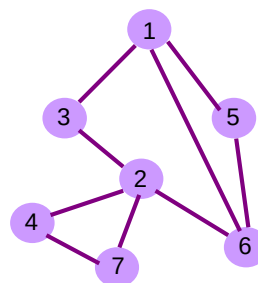


Steps in Spanning Tree Algorithm

- Initially, each switch thinks it is the root
 - Switch sends a message out every interface
 - ... identifying itself as the root with distance 0
 - Example: switch X announces (X, 0, X)
- Switches update their view of the root
 - Upon receiving a message, check the root id
 - If the new id is smaller, start viewing that switch as root
- Switches compute their distance from the root
 - Add 1 to the distance received from a neighbor
 - Identify interfaces not on a shortest path to the root
 - ... and exclude them from the spanning tree

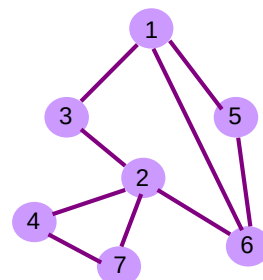
Example From Switch #4's Viewpoint

- Switch #4 thinks it is the root
 - Sends (4, 0, 4) message to 2 and 7
- Then, switch #4 hears from #2
 - Receives (2, 0, 2) message from 2
 - ... and thinks that #2 is the root
 - And realizes it is just one hop away
- Then, switch #4 hears from #7
 - Receives (2, 1, 7) from 7
 - And realizes this is a longer path
 - So, prefers its own one-hop path
 - And removes 4-7 link from the tree



Example From Switch #4's Viewpoint

- Switch #2 hears about switch #1
 - Switch 2 hears (1, 1, 3) from 3
 - Switch 2 starts treating 1 as root
 - And sends (1, 2, 2) to neighbors
- Switch #4 hears from switch #2
 - Switch 4 starts treating 1 as root
 - And sends (1, 3, 4) to neighbors
- Switch #4 hears from switch #7
 - Switch 4 receives (1, 3, 7) from 7
 - And realizes this is a longer path
 - So, prefers its own three-hop path
 - And removes 4-7 link from the tree



Robust Spanning Tree Algorithm

- Algorithm must react to failures
 - Failure of the root node
 - Need to elect a new root, with the next lowest identifier
 - Failure of other switches and links
 - Need to recompute the spanning tree
- Root switch continues sending messages
 - Periodically reannouncing itself as the root (1, 0, 1)
 - Other switches continue forwarding messages
- Detecting failures through timeout (soft state!)
 - Switch waits to hear from others
 - Eventually times out and claims to be the root

What are routers?

- Routers are another type of internetworking device and they connect two or more packet-switched networks or subnetworks.
- These devices pass data packets between networks based on network protocol or layer 3 information.
- Routers have the ability to make intelligent decisions as to the best path for delivery of data on the network.

How do IP addresses differ from MAC addresses?

- Like MAC addresses, every IP address is unique. No two IP addresses are ever alike.
- However, while MAC addresses are physical addresses that are actually hard-coded into the NIC card and occur at the data link layer
- IP addresses are implemented in software and occur at the network layer of the OSI model.

How do routers work?

- A router helps direct data packets to their destination IP address.
- Routers use an internal routing table - a list of paths to various network destinations – to route packets to IP destination.
- The router reads a packet's header to determine where it is going, then consults the routing table to figure out the most efficient path to that destination; it then forwards the packet to the next network in the path.

Functions of router

- Control plane
 - Control plane is responsible for populating the routing table, drawing network topology, forwarding table and hence enabling the data plane functions.
- Data plane
 - Routing table, forwarding table and the routing logic constitute the data plane function. Data plane packet goes through the router and incoming and outgoing of packets are done based on control plane logic.



Types of routers

- *Core router*: Unlike the routers used within a home or small business LAN, a core router is used by large corporations and businesses that transmit a high volume of data packets within their network.
- *Edge router*: An edge router communicates with both core routers and external networks. Edge routers live at the "edge" of a network and use the **BGP (Border Gateway Protocol)** to send and receive data from other LANs and WANs.
- *Virtual router*: A virtual router is a software application that performs the same function as a standard hardware router.

